Project Proposal

Here is the proposal template, which must be validated your lecturer before starting the development.
Replace grey text with your own information.
File Format: **GroupNumber_TeamNumber_Project_Proposal.pdf**
Example: G5_T1_Project_Proposal.pdf

Project Title

The TitanLift

Project Type

[ ] Game Project
[X] Resource Management Project

Team Members

Choeun Linda
Oeurn Meylih

Project Description

We are building a *smart elevator system* for 10 floors building that features a rooftop sky bar and an exclusive VIP floor (Floor 9). The system operates as a standard elevator for general passengers while incorporating a VIP express mode triggered by a secure card scan. When activated, the elevator bypasses all pending general floor requests and travels directly to Floor 9 without stopping. Access to Floor 9 is fully restricted to verified cardholders only.

Project Data

The system will use three main structs — Floor, Request, and Elevator — to keep track of floor details, passenger requests, and the elevator's current state. All floors from Ground to Rooftop will be stored in an array of 12 entries. Passenger requests will be managed through a waiting list that supports VIP priority, and a separate list will store valid VIP card IDs for verification. Fixed values like floor numbers and limits will be defined as constants. Once VIP express mode is turned on, the elevator will skip all regular floor requests in both directions until the VIP passenger reaches their destination.

Project Features

List at least 4 core features your project will include.

- Standard elevator service that handles all general floor requests using a normal up-and-down movement.
- VIP express mode triggered by a card scan — elevator skips all general requests and goes directly to Floor 9.
- Card scan access control that restricts Floor 9 to verified cardholders only, with scanners at 2locations (*Inside the cabin and Floor 9 landing*).
- Priority request system that holds all general requests while express mode is active and resumes normal service after VIP delivery.

Team Contribution Plan

Who is doing what? How will you divide the tasks?

- Linda: Elevator movement, request queue, SCAN algorithm, and express mode logic.
- Meylih: Floor and building setup, card scan authentication, floor locking, and display output.

Additional information

Reference : Claude AI (Anthropic, 2025). Available at: https://claude.ai
Concept inspired by :The leaf coffee (Noden2 Hotel & Apartment) Noden2